

Required Test Points

Power

- **+BATT_IN:** TP102
- **+CHRG:** TP204 / TP206 / TP208
- **GND:** TP103 / TP104 / TP203 / TP205

ADC

- **+BATT_SYS:** TP201
- **+SYS:** TP202 / TP207
- **+3.3V:** TP301
- **+VBCKP:** TP607

UART1 (nRF9160 debug)

- **DUT.RX:** TP401 (data into nRF91)
- **DUT.TX:** TP402 (data out from nRF91)

UART2 (nRF52840 debug)

- **DUT.RX:** TP502 (data into nRF52)
- **DUT.TX:** TP501 (data out from nRF52)

J-Link1 (nRF9160)

- **SWDIO:** TP407
- **SWDCLK:** TP408
- **RESET:** TP409

J-Link2 (nRF52840)

- **SWDIO:** TP503
- **SWDCLK:** TP504
- **RESET:** TP514

I2C (nRF52840)

- **SCL:** TP511
- **SDA:** TP512

Electrical Power Test

Objectives

- Ensure board can receive power at battery input terminal.
- Ensure battery undervoltage lockout cuts off power below 3.5V.
- Ensure charger load sharing works.

1. Apply +3.4V to **+BATT_IN**
2. Ensure device electrical state.
 - a. Ensure **+BATT_SYS** voltage is below 0.3V.
 - b. Ensure **+3.3V** voltage is below 0.3V.
 - c. Ensure **+VBCKP** voltage is below 0.3V.
 - d. Ensure current consumption is below 1uA.
3. Apply +3.7V to **+BATT_IN**
4. Ensure **+3.3V** remains below 0.3V for 2 seconds.
5. Ensure device electrical state.
 - a. Ensure **+BATT_SYS** voltage is same as **+BATT_IN**
 - b. Ensure **+SYS** voltage is same as **+BATT_IN**
 - c. Ensure **+3.3V** voltage is within 3.2V – 3.4V.
 - d. Ensure **+VBCKP** voltage is within 2.4V – 2.6V.
 - e. Ensure proper current consumption (no short circuits)
6. Apply +4.5V to **+BATT_IN**
7. Ensure device electrical state.
 - a. Ensure **+SYS** voltage is same as **+BATT_IN**
8. Apply 5.0V to **+CHRG**
9. Ensure device electrical state.
 - a. Ensure **+SYS** voltage is 5V.
 - b. Ensure **+3.3V** voltage is within 3.2V – 3.4V.
 - c. Ensure **+VBCKP** voltage is within 2.4V – 2.6V.
10. Remove power from **+CHRG**

Flash Manufacturing Firmware

Objectives

- Ensure both microcontrollers are programmed with correct firmware for testing.
1. Flash nRF9151 modem firmware
 2. In parallel, flash nRF52840 with manufacturing firmware.
 3. When modem firmware is complete, flash nRF9151 with manufacturing firmware.

Execute Test Firmware

Objectives

- Communicate via UART with nRF9151 manufacturing firmware to execute various tests.
1. Ensure the board responds to NOP command.
 2. Query board for its battery voltage reading.
 3. Ensure voltage reading is appropriate for the **+BATT_IN** voltage.
 4. Send POST command.
 5. Ensure POST pass criteria.
 - a. Verify correct chip id for all sensors, ICs, and modems.
 - b. Read, modify, write, read test for external flash.
 6. Read IMEI and all ICCIDs.
 7. Read accelerometer X/Y/Z and temperature values.
 - a. Ensure values are correct by comparing them to reference values from MTIB.
 8. Send hard reset command via UART.
 - a. Monitor **+3.3V** and ensure it drops below 1.8V.
 - b. Monitor **+VBCKP** and ensure it remains at 2.4-2.6V.

Personalize Device

Objectives

- Personalize device via UART.
1. If test firmware passed, personalize the device with data obtained from CoreOps.